

A Polynomial Distance-Multiplicity Gap for Every Large Cardinality

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July 30, 2026

Abstract

For a finite set $A \subset \mathbb{R}^2$, let $f_A(r)$ be the number of unordered pairs at distance r , and let $G(n)$ be the maximum, over all n -point sets, of the gap between the two largest values of f_A . We prove that $G(n) \geq n^{1+c}$ for every sufficiently large n , for an absolute effective constant $c > 0$. The proof, found by GPT 5.6 Sol, builds on a recent OpenAI advance on the unit-distance problem [2], reusing part of its arithmetic construction.

1 Statement and tower input

For a finite planar set A , list the values $f_A(r)$, over the distinct positive distances determined by A , as

$$a_1(A) \geq a_2(A) \geq \cdots,$$

with zeros appended when necessary, and define

$$\text{gap}(A) = a_1(A) - a_2(A), \quad G(n) = \max_{|A|=n} \text{gap}(A).$$

This is Erdős Problem #959 [1].

Theorem 1.1. *There is an effective absolute constant $c > 0$ such that*

$$G(n) \geq n^{1+c}$$

for every sufficiently large integer n .

We use the arithmetic construction of [2, Proposition 3.8 and its proof] only through the following consequence.

Proposition 1.2 (OpenAI tower consequence). *There exist constants $R, H > 1$, a rational prime $q \equiv 1 \pmod{4}$, a positive integer d_0 , and totally real fields*

$$F_0 \subset F_1 \subset F_2 \subset \cdots$$

such that, writing $d_j = [F_j : \mathbb{Q}]$ and $K_j = F_j(i)$,

- (i) $d_j = d_0 3^j$;*
- (ii) q splits completely in every F_j ;*
- (iii) $\text{rd}(K_j) \leq R$ and $h(K_j) \leq H^{d_j}$.*

All constants may be chosen effectively.

Proof. The cited construction gives an infinite unramified pro-3 extension of a fixed totally real field in which finitely many rational primes $q \equiv 1 \pmod{4}$ split completely, and it gives uniform root-discriminant and exponential class-number bounds after adjoining i . Choose one such q . Inside the infinite pro-3 Galois group, recursively take a subgroup of index 3 in the preceding open subgroup. The corresponding finite fields are nested and have successive degree ratio 3. Complete splitting, the root-discriminant bound, and the class-number bound pass to these layers. $\square$

Fix the constants and one tower from Proposition 1.2. Throughout the next three sections, F is one tower field, $d = [F : \mathbb{Q}]$, $K = F(i)$, and σ is complex conjugation over F .

The norm-class group $\mathcal{G}_K$ consists of pairs (J, u) , where J is a fractional ideal of K and $u \in F^\times$ generates $N_{K/F}(J)$, modulo $(J, u) \sim ((a)J, a\sigma(a)u)$. It fits into the standard exact sequence

$$\mathcal{O}_K^\times \xrightarrow{N_{K/F}} \mathcal{O}_F^\times \longrightarrow \mathcal{G}_K \longrightarrow \text{Cl}(K) \xrightarrow{N_{K/F}} \text{Cl}(F).$$

Since $(\mathcal{O}_F^\times)^2 \subset N_{K/F}(\mathcal{O}_K^\times)$ and $|\mathcal{O}_F^\times/(\mathcal{O}_F^\times)^2| = 2^d$, Proposition 1.2 gives

$$|\mathcal{G}_K| \leq 2^d h(K) \leq e^{\kappa d}, \quad \kappa = \log(2H). \quad (1)$$

2 An isolated relative-norm shell

Choose a positive rational number $\eta = a/b$ in lowest terms, with $a \nmid b$, and put

$$s = \frac{2\eta}{\pi}, \quad k_\eta(Q) = \left\lfloor \frac{1}{Q^\eta - 1} \right\rfloor, \quad T = \frac{2}{\sqrt{\eta}}. \quad (2)$$

The value of η will be fixed sufficiently small in Lemma 2.3.

Define

$$\mathfrak{A} = \prod_{\substack{\mathfrak{p} \subset \mathcal{O}_F \\ \mathfrak{p} \text{ split in } K/F}} \mathfrak{p}^{k_\eta(N\mathfrak{p})}. \quad (3)$$

Only primes of norm at most $2^{1/\eta}$ occur. For each prime in this product, choose one of the two primes of K above it, with the prescribed exponent, and let J_0 be their product. Thus $N_{K/F}(J_0) = \mathfrak{A}$.

For an integral ideal $\mathfrak{B} \subset \mathcal{O}_F$, set

$$a_K(\mathfrak{B}) = \#\{J \subset \mathcal{O}_K : J \text{ integral and } N_{K/F}(J) = \mathfrak{B}\}.$$

The ideals of relative norm $\mathfrak{A}$ occupy certain classes $\mathcal{C}(\mathfrak{A}) \subset \mathcal{G}_K$; write $m = |\mathcal{C}(\mathfrak{A})|$. For each occupied class g , choose a representative (J_g, u_g) with u_g totally positive and set

$$I_g = J_g^{-1} J_0^{-1}, \quad \alpha_g = u_g^{-1}.$$

Then $N_{K/F}(I_g) = \mathfrak{A}^{-1}(\alpha_g)$. For totally positive $\gamma \in F^\times$, define

$$\mathcal{S}_g(\gamma) = \{\beta \in I_g : \beta\sigma(\beta) = \gamma\alpha_g\}.$$

Proposition 2.1 (Shell count). *If some $\mathcal{S}_g(\gamma)$ is nonempty, then $\mathfrak{B} = (\gamma)\mathfrak{A}$ is integral and*

$$\sum_g |\mathcal{S}_g(\gamma)| \leq w_K a_K(\mathfrak{B}),$$

where w_K is the number of roots of unity in K . For the target shell,

$$\sum_g |\mathcal{S}_g(1)| = w_K a_K(\mathfrak{A}).$$

Proof. For $\beta \in \mathcal{S}_g(\gamma)$, the ideal $J = (\beta)J_gJ_0$ is integral of relative norm $(\gamma)\mathfrak{A}$, and $[JJ_0^{-1}, \gamma] = g$. Conversely, every ideal of that relative norm in an occupied class arises in this way. Two solutions produce the same ideal exactly when their quotient is a norm-one unit; in a CM extension these form the finite group of roots of unity. For $\gamma = 1$, every ideal of relative norm $\mathfrak{A}$ is occupied. $\square$

Proposition 2.2 (Multiplicative isolation). *There is a constant $\rho_\eta \in (0, 1)$ such that, for every integral $\mathfrak{B} \neq \mathfrak{A}$,*

$$\frac{a_K(\mathfrak{B})}{a_K(\mathfrak{A})} \leq \rho_\eta \left(\frac{N\mathfrak{B}}{N\mathfrak{A}} \right)^\eta. \quad (4)$$

Moreover,

$$a_K(\mathfrak{A}) \geq N(\mathfrak{A})^\eta. \quad (5)$$

If

$$\nu = \frac{1}{d} \log N(\mathfrak{A}), \quad \alpha = \frac{1}{d} \log a_K(\mathfrak{A}),$$

then

$$k_\eta(q) \log q \leq \nu \leq C_\eta, \quad \alpha \geq \log(k_\eta(q) + 1), \quad (6)$$

where

$$C_\eta = \sum_{p \leq 2^{1/\eta}} \max_{\substack{f \geq 1 \\ p^f \leq 2^{1/\eta}}} k_\eta(p^f) \log p.$$

Proof. The function a_K is multiplicative. At a split prime of norm Q , $a_K(\mathfrak{p}^e) = e + 1$, and the weighted sequence

$$(e + 1)Q^{-\eta e}$$

has its maximum at $e = k_\eta(Q)$. The maximum is unique. Indeed, equality of two consecutive values would imply

$$Q^a(e + 1)^b = (e + 2)^b.$$

Writing $Q = p^f$ and using $\gcd(e + 1, e + 2) = 1$ forces $e = 0$, $p = 2$, and $af = b$, contrary to $a \nmid b$. Hence the ratio of the second-largest local weighted value to the largest is uniformly below one; only finitely many prime-power norms need be checked.

At an inert prime, $a_K(\mathfrak{p}^e)$ is 1 for even e and 0 for odd e , while at a ramified prime it is 1 for every e . The target exponent is zero in both cases, and every nontrivial weighted deviation is at most $2^{-2\eta}$ or $2^{-\eta}$. Multiplication of the local inequalities proves (4). Comparing the maximizing split exponent with $e = 0$ proves (5).

For the upper bound on ν , write $p\mathcal{O}_F = \prod_i \mathfrak{p}_i^{e_i}$ with residue degrees f_i . Since $\sum_i e_i f_i = d$, the contribution of the primes over p is at most

$$d \max_{f: p^f \leq 2^{1/\eta}} k_\eta(p^f) \log p.$$

Summing over p gives $\nu \leq C_\eta$. Finally, q splits completely in F and, because $q \equiv 1 \pmod{4}$, every prime above q splits in K/F . The resulting d local factors give the two lower bounds in (6). $\square$

The next lemma fixes the only parameter choice in the proof.

Lemma 2.3 (Choice of η). *The rational number $\eta = a/b$ in (2) may be chosen, effectively and independently of the tower level, so that, with*

$$V_\eta = k_\eta(q) \log q, \quad A_\eta = \log(k_\eta(q) + 1),$$

one has

$$A_\eta - \kappa - \log 2 - \eta > 0, \quad (7)$$

$$\log 2 + V_\eta - \log R - \log s > 0, \quad (8)$$

$$\Psi_\eta > 0. \quad (9)$$

where

$$\begin{aligned} \Psi_\eta &= 2(\log 2 + V_\eta - \log R) - 2\log(2s) - 2\eta - \kappa \\ &\quad - 2\log(1 + 2Te^{V_\eta/2}) - 2\eta\log(4T^2). \end{aligned}$$

It may also be chosen so that the Poisson-summation errors used below are $O(e^{-10d})$ uniformly in the tower.

Proof. As $\eta \downarrow 0$,

$$V_\eta \sim \eta^{-1}, \quad A_\eta = \log(\eta^{-1}) + O_q(1).$$

Thus (7) and (8) hold for sufficiently small η . Since $T = 2\eta^{-1/2}$,

$$\Psi_\eta = V_\eta + O_{R,H,q}(\log(\eta^{-1})) \longrightarrow \infty.$$

The dual-lattice estimate below has exponential coefficient at least $\pi^2 e^{V_\eta}/(\eta R^2)$, which also tends to infinity. Rational numbers a/b with $a \nmid b$ occur arbitrarily close to zero, so all conditions can be met simultaneously and checked effectively. $\square$

3 Gaussian blocks

Choose one complex embedding of K above each real embedding $\tau_j : F \hookrightarrow \mathbb{R}$ and define

$$\iota_g(\beta) = \left(\frac{\tau_j(\beta)}{\sqrt{\tau_j(\alpha_g)}} \right)_{j=1}^d, \quad \Lambda_g = \iota_g(I_g) \subset \mathbb{C}^d.$$

For $\beta \in \mathcal{S}_g(\gamma)$,

$$|\iota_g(\beta)_j|^2 = \tau_j(\gamma), \quad \|\iota_g(\beta)\|^2 = \text{Tr}_{F/\mathbb{Q}}(\gamma). \quad (10)$$

In particular, the target vectors have squared length d .

The standard ideal-lattice calculation gives

$$\text{covol}(\Lambda_g) = D := 2^{-d} \frac{\sqrt{|\Delta_K|}}{N(\mathfrak{A})}, \quad (11)$$

and, for nonzero $v \in \Lambda_g$ and $\xi \in \Lambda_g^*$,

$$\|v\|^2 \geq de^{-\nu}, \quad \|\xi\|^2 \geq 4d \frac{e^\nu}{\text{rd}(K)^2}. \quad (12)$$

These are the usual norm-product bounds for an ideal lattice and its trace dual.

By Equations (6) and (12) and Lemma 2.3, Poisson summation gives, uniformly in g , $u \in \mathbb{C}^d$, and $a \in \{s/2, s, 3s/2, 2s\}$,

$$\sum_{x \in \Lambda_g} e^{-\pi a \|x+u\|^2} = \frac{a^{-d}}{D} (1 + O(e^{-10d})). \quad (13)$$

Completing the square gives, for $v \in \Lambda_g$,

$$\sum_{x \in \Lambda_g} e^{-\pi s(\|x\|^2 + \|x+v\|^2)} = \frac{(2s)^{-d}}{D} e^{-\pi s\|v\|^2/2} (1 + O(e^{-10d})). \quad (14)$$

Retain each $x \in \Lambda_g$ independently with probability $e^{-\pi s\|x\|^2}$. Let X be the total number retained, and let Y_γ be the number of ordered retained pairs whose displacement lies in $\iota_g(\mathcal{S}_g(\gamma))$, summed over g . Then

$$\mathbb{E}X = m \frac{s^{-d}}{D} (1 + O(e^{-10d})), \quad (15)$$

$$\mathbb{E}Y_1 = \frac{(2s)^{-d}}{D} e^{-\eta d} w_K a_K(\mathfrak{A}) (1 + O(e^{-10d})). \quad (16)$$

Proposition 3.1 (Expected shell separation). *There is $\delta > 0$, independent of the tower level, such that for every $\gamma \neq 1$ and all sufficiently large d ,*

$$\mathbb{E}(Y_1 - Y_\gamma) \geq \delta \mathbb{E}Y_1.$$

Proof. If $\mathfrak{B} = (\gamma)\mathfrak{A} \neq \mathfrak{A}$, then Propositions 2.1 and 2.2 and Equation (14) give

$$\frac{\mathbb{E}Y_\gamma}{\mathbb{E}Y_1} \leq \rho_\eta \exp \left(-\eta \sum_{j=1}^d [\tau_j(\gamma) - 1 - \log \tau_j(\gamma)] \right) + O(e^{-10d}) \leq \rho_\eta + O(e^{-10d}).$$

If $\mathfrak{B} = \mathfrak{A}$ and $\gamma \neq 1$, then γ is a nontrivial totally positive unit. Since $|\mathrm{N}_{F/\mathbb{Q}}(\gamma - 1)| \geq 1$, some conjugate is at least 2. AM–GM on the remaining conjugates yields

$$\mathrm{Tr}_{F/\mathbb{Q}}(\gamma) \geq d + 1 - \log 2,$$

so

$$\frac{\mathbb{E}Y_\gamma}{\mathbb{E}Y_1} \leq e^{-\eta(1-\log 2)} + O(e^{-10d}).$$

Take δ smaller than half of $1 - \max\{\rho_\eta, e^{-\eta(1-\log 2)}\}$. □

We now restrict each lattice to the finite ball

$$\mathcal{Q}_g = \{x \in \Lambda_g : \|x\| \leq T\sqrt{d}\}.$$

Using Equation (13) with the Gaussian weight split into two factors, the relative point-mass loss is

$$O\left(e^{-(\eta T^2 - \log 2)d}\right) = O(e^{-3d}),$$

and, for target pairs, completing the square gives relative loss

$$O\left((4/3)^d e^{-\eta(T^2 - 1/3)d}\right) = O(e^{-3d}).$$

Thus Proposition 3.1 remains valid after truncation.

Let $M = \sum_g |\mathcal{Q}_g|$. By Equation (12) and packing,

$$M \leq m(1 + 2Te^{\nu/2})^{2d} = e^{O_\eta(d)}. \quad (17)$$

There are at most M^2 candidate displacement shells. Every candidate γ satisfies $\mathrm{N}_F(\gamma) \leq (4T^2)^d$. Changing one vertex indicator therefore changes $Y_1 - Y_\gamma$ by at most

$$4w_K a_K(\mathfrak{A})(4T^2)^{\eta d}. \quad (18)$$

Proposition 3.2 (Simultaneous finite extraction). *For every sufficiently large tower degree there is a realization of the truncated sample for which*

$$\frac{1}{2}\mathbb{E}X \leq X \leq 2\mathbb{E}X, \quad Y_1 \geq \frac{1}{2}\mathbb{E}Y_1,$$

and

$$Y_1 - Y_\gamma \geq \frac{\delta}{2}\mathbb{E}Y_1 \quad (\gamma \neq 1)$$

simultaneously for every candidate shell.

Proof. For a fixed competitor, McDiarmid's inequality [3] and Equation (18) bound the failure probability by

$$\exp\left(-\frac{(\mathbb{E}(Y_1 - Y_\gamma))^2}{32Mw_K^2a_K(\mathfrak{A})^2(4T^2)^{2\eta d}}\right).$$

After inserting Equations (15) to (17), the logarithm of the quantity in the outer exponential, divided by d , is at least

$$\Psi_\eta + o(1) > 0$$

by Lemma 2.3; monotonicity in ν allows the lower bound $\nu \geq V_\eta$. Hence each shell fails with probability $\exp(-e^{cd})$ for some $c > 0$, whereas the number of shells is only $e^{O(d)}$. The same estimate applies to Y_1 , and Chernoff's inequality applies to X ; a union bound leaves positive probability. $\square$

4 Planar blocks and all cardinalities

Project each nonempty sampled component to its first complex coordinate. This projection is injective because a field element with one zero embedding is zero. Distinct shell parameters give distinct projected squared distances. Place the finitely many projected components by generic orientation-preserving rigid motions so that every cross-component distance is new and has multiplicity one; this follows by avoiding finitely many algebraic curves in the motion parameters.

Proposition 4.1 (Uniform blocks). *There are effective constants $p_+, \ell_- > 0$ such that every sufficiently large tower degree d produces a planar set A_d satisfying*

$$|A_d| \leq e^{p_+d}, \quad \text{gap}(A_d) \geq |A_d|e^{\ell_-d}.$$

Proof. By Equations (1), (6), (11) and (15) and $\text{rd}(K) \geq 1$, we may take

$$p_+ = \kappa + \log \frac{\pi}{\eta} + C_\eta + 1.$$

Passing from ordered to unordered pairs divides every internal shell count by 2. By Proposition 3.2 and Equations (15) and (16), the gap between the target shell and every other internal shell is, up to a fixed factor,

$$|A_d| 2^{-d} e^{-\eta d} \frac{w_K a_K(\mathfrak{A})}{m}.$$

Using Equations (1), (6) and (7), this is at least $|A_d|e^{(L+o(1))d}$, where

$$L = A_\eta - \kappa - \log 2 - \eta > 0.$$

Cross-component multiplicities are one, so for large d they do not interfere. Take $\ell_- = L/2$. $\square$

Proof of Theorem 1.1. Let p_+, ℓ_- be as above and set

$$\theta = \frac{1}{4p_+}, \quad c = \frac{\ell_-}{24p_+}.$$

The available degrees are $d_j = d_0 3^j$. Hence, for every sufficiently large n , one may choose an available d with

$$\frac{\theta}{3} \log n \leq d \leq \theta \log n.$$

For the corresponding block, $N = |A_d| \leq n^{1/4}$. Take $t = \lfloor n/N \rfloor$ congruent copies and place them generically. Internal multiplicities are multiplied by t , while every cross-copy distance has multiplicity one. Therefore

$$\text{gap}(B) \geq t \text{gap}(A_d) - 1 \geq \frac{n}{3} e^{\ell_- d} \geq \frac{1}{3} n^{1+\ell_-/(12p_+)}$$

for all sufficiently large n .

Add the remaining points one at a time, each time avoiding the finitely many points, circles, and perpendicular bisectors that would repeat an existing or new distance. All added distances then have multiplicity one, so the two largest multiplicities and their gap are unchanged. Absorbing the factor $1/3$ gives $G(n) \geq n^{1+c}$. $\square$

References

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- [3] C. McDiarmid, On the method of bounded differences, in *Surveys in Combinatorics 1989*, London Math. Soc. Lecture Note Series 141, Cambridge University Press, 1989, 148–188.